

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the source code.	Yes	Improves readability and follows Python coding standards.
2	Proper File Structure	Project files and folders are organized logically.	Yes	Separate folders for templates, static files, model, dataset, and application code.
3	Meaningful Variable Names	Variables clearly represent their purpose.	Yes	Descriptive names such as applicant_data, prediction_result, and model.
4	Function / Method Names	Functions are named according to the task they perform.	Yes	Functions such as preprocess_data(), predict_approval(), and load_model() improve code understanding.
5	Code Comments	Important sections include comments and documentation.	Yes	Comments explain preprocessing steps, model loading, and prediction logic.
6	Modular Design	Code is divided into reusable functions and modules.	Yes	Frontend, backend, model loading, and preprocessing are implemented as separate components.
7	No Redundant Code	Duplicate and unused code has been removed.	Yes	Common functionality is reused to avoid repetition.

8	Error Handling	Exceptions and invalid inputs are handled appropriately.	Yes	Input validation and exception handling prevent application crashes.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	Model training and prediction pipeline	High
2	Machine Learning Prediction Module	Python, Scikit-learn	Used whenever a prediction request is made	High
3	Flask Routing Module	Python, Flask	Handles navigation between all web pages	High
4	Frontend UI Components	HTML5, CSS3, JavaScript, Bootstrap	Shared across Home, Prediction, About, and Dashboard pages	Medium
5	Model Loading Module	Python, Joblib	Loads the trained model for every prediction request	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	The project follows a clean folder hierarchy with well-organized modules and files.
Readability	5	Meaningful variable names, consistent formatting, and comments make the code easy to understand.
Reusability	5	Core functionality is modular, allowing components such as preprocessing and prediction to be reused.
Documentation / Comments	4	Important logic is documented with comments; additional API documentation could further improve maintainability.
Overall Score	4.8 / 5	The codebase is clean, modular, maintainable, and aligns with good software development practices.